

### Coding & Solution

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 7 July 2026                     |
| <b>Team ID</b>       | XXXXXX                          |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

#### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

### Solution Summary

| Field                        | Details                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b> | <a href="https://github.com/jahnavikaranam17/Credit-Card-Approval-Prediction">https://github.com/jahnavikaranam17/Credit-Card-Approval-Prediction</a> |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Language(s)</b>       | Python                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Framework(s) Used</b>             | Flask                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Key Features Implemented</b>      | <ul style="list-style-type: none"> <li>• Data preprocessing and cleaning</li> <li>• Feature encoding and normalization</li> <li>• Credit card approval prediction using Machine Learning</li> <li>• Random Forest classification model</li> <li>• Applicant information input form</li> <li>• Instant approval/rejection prediction</li> <li>• Accuracy evaluation using performance metrics</li> </ul> |
| <b>Pending / Incomplete Features</b> | <ul style="list-style-type: none"> <li>• User authentication</li> <li>• Cloud deployment</li> <li>• Admin dashboard</li> <li>• Model retraining with live data</li> </ul>                                                                                                                                                                                                                               |
| <b>Setup / Run Instructions</b>      | <ol style="list-style-type: none"> <li>1. Clone the GitHub repository.</li> <li>2. Install the required Python libraries.</li> <li>3. Run app.py using Flask.</li> <li>4. Open the application in a web browser.</li> <li>5. Enter applicant details.</li> <li>6. Click Predict to view the approval result.</li> </ol>                                                                                 |



### Code Quality Checklist

| S.N o | Criteria                                               | Status (Yes / No) |
|-------|--------------------------------------------------------|-------------------|
| 1     | Code is modular and organized into functions / classes | Yes               |
| 2     | Meaningful variable and function names are used        | Yes               |
| 3     | Code includes comments / documentation where necessary | Yes               |

|   |                                                       |     |
|---|-------------------------------------------------------|-----|
| 4 | Error handling is implemented for critical operations | Yes |
| 5 | The application runs without critical errors          | Yes |
| 6 | Code is committed to a version control repository     | Yes |

**Additional Notes / Comments:**

The project follows clean coding practices with a modular architecture. Data preprocessing, model training, prediction, and web interface are implemented as separate components, making the application easy to maintain, test, and extend.